Version 7.2.12

PVsyst - Simulation report

Grid-Connected System

Project: Training4

Variant: New simulation variant

Sheds, single array

System power: 10.50 kWp

New Capital - Egypt

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04/07/22 14:53
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Project summary

Geographical Site
New Capital
Egypt

Situation

Latitude	29.94 °N
Longitude	31.80 °E
Altitude	389 m
Time zone	UTC+2

Project settings
Albedo 0.20

Meteo data
New Capital
Meteonorm 7.1 (1981-2009), Sat=74% - Synthetic

System summary

Grid-Connected System

Sheds, single array

PV Field Orientation
Fixed plane
Tilt/Azimuth 30 / 0 °

Near Shadings
Linear shadings

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	28 units
Pnom total	10.50 kWp

Inverters

Nb. of units	1 unit
Pnom total	10.00 kWac
Pnom ratio	1.050

Results summary

Produced Energy	18.85 MWh/year	Specific production	1795 kWh/kWp/year	Perf. Ratio PR	78.46 %

Table of contents

Project and results summary 2
General parameters, PV Array Characteristics, System losses 3
Near shading definition - Iso-shadings diagram 5
Main results 6
Loss diagram 7
Special graphs 8

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VC0, Simulation date:
04/07/22 14:53
with v7.2.12

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General parameters

Grid-Connected System

Sheds, single array

PV Field Orientation

Orientation	
Fixed plane	
Tilt/Azimuth	30 / 0 °

Sheds configuration	
Nb. of sheds	2 units
Single array	
Sizes	
Sheds spacing	7.00 m
Collector width	3.94 m
Ground Cov. Ratio (GCR)	56.3 %
Top inactive band	0.02 m
Bottom inactive band	0.02 m
Shading limit angle	
Limit profile angle	29.0 °

Models used	
Transposition	Perez
Diffuse	Perez, Meteonorm
Circumsolar	with diffuse

Horizon
Free Horizon

Near Shadings
Linear shadings

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	ARECO
Model	ARECO M375 – MonoPerc
(Custom parameters definition)	
Unit Nom. Power	375 Wp
Number of PV modules	28 units
Nominal (STC)	10.50 kWp
Modules	2 Strings x 14 In series
At operating cond. (50°C)	
Pmpp	9.54 kWp
U mpp	510 V
I mpp	19 A

Inverter

Manufacturer	Kaco new energy
Model	Blueplanet 10.0 TL3
(Original PVsyst database)	
Unit Nom. Power	10.00 kWac
Number of inverters	1 unit
Total power	10.0 kWac
Operating voltage	200-800 V
Pnom ratio (DC:AC)	1.05

Total PV power

Nominal (STC)	11 kWp
Total	28 modules
Module area	54.4 m²
Cell area	49.3 m²

Total inverter power

Total power	10 kWac
Number of inverters	1 unit
Pnom ratio	1.05

Array losses

Array Soiling Losses
Loss Fraction 5.0 %

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 29.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 453 mΩ
Loss Fraction 1.5 % at STC

LID - Light Induced Degradation
Loss Fraction 2.0 %

Module Quality Loss
Loss Fraction -0.8 %

Module mismatch losses
Loss Fraction 2.0 % at MPP

Strings Mismatch loss
Loss Fraction 0.1 %

IAM loss factor
ASHRAE Param: IAM = 1 - bo(1/cosi -1)
bo Param. 0.03

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AC wiring losses

Inv. output line up to injection point

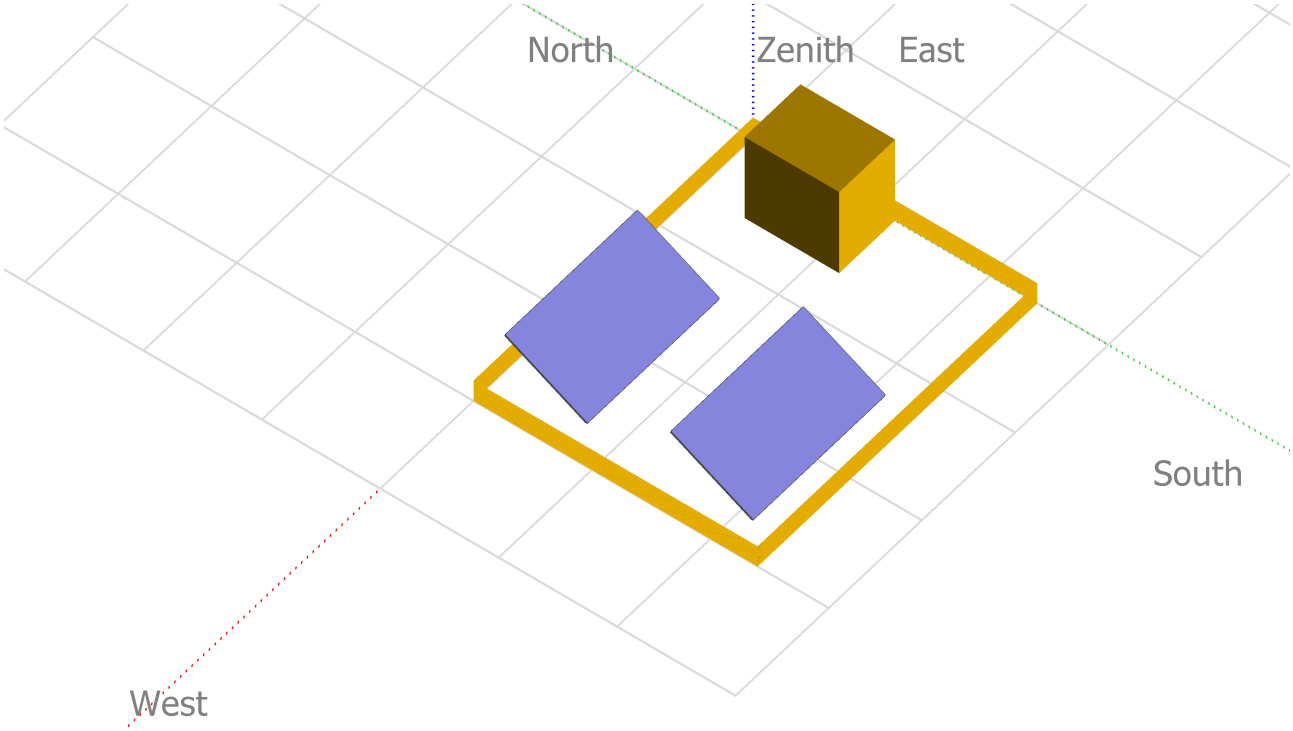
Inverter voltage	400 Vac tri
Loss Fraction	1.00 % at STC

Inverter: Blueplanet 10.0 TL3

Wire section (1 Inv.)	Copper 1 x 3 x 6 mm²
Wires length	50 m

Near shadings parameter

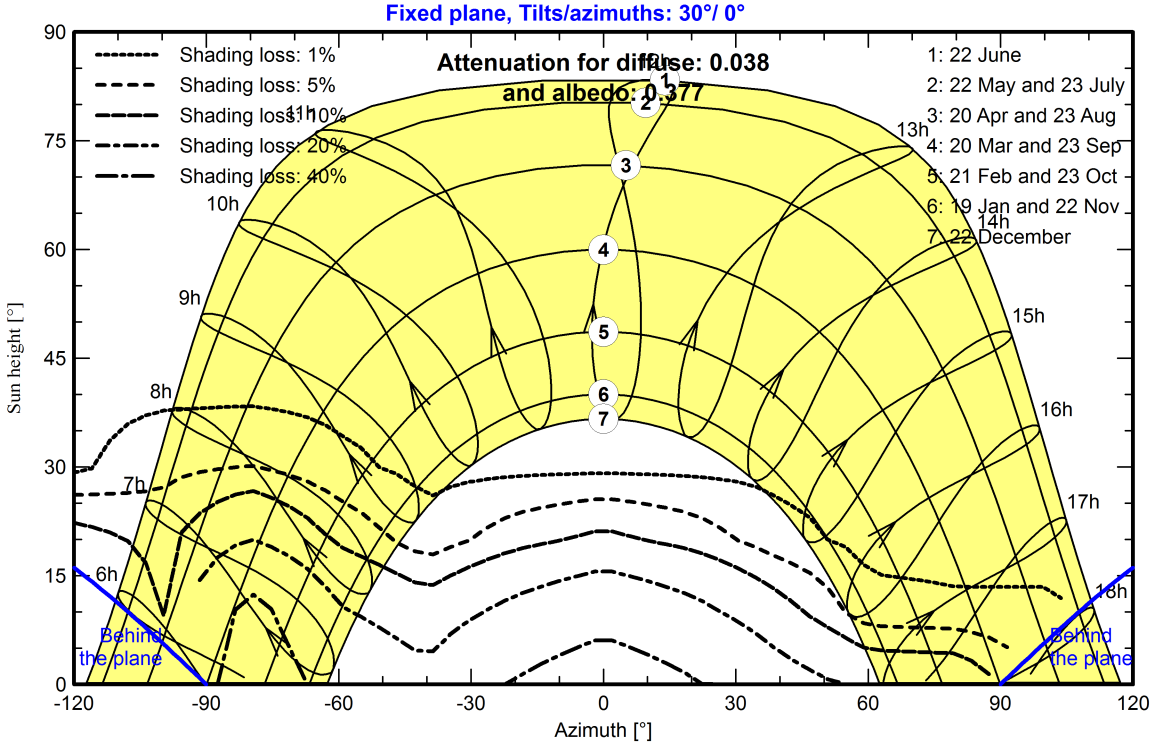
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 30°/ 0°

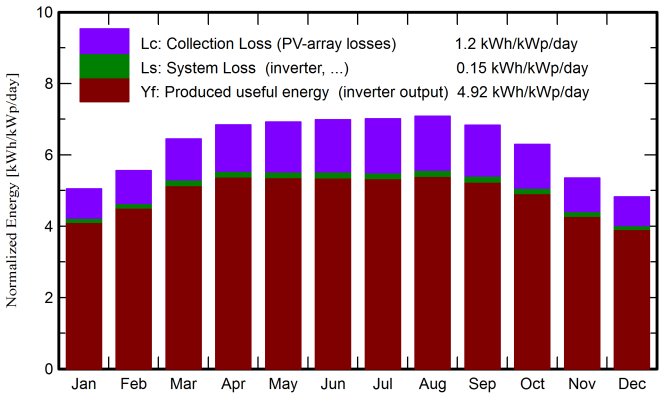


Main results

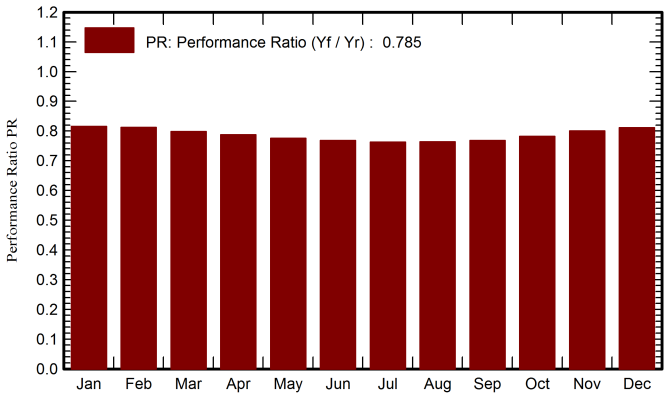
System Production

Produced Energy	18.85 MWh/year	Specific production	1795 kWh/kWp/year
		Performance Ratio PR	78.46 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray MWh	E_Grid MWh	PR ratio
January	105.6	37.64	12.73	156.3	142.5	1.377	1.338	0.815
February	118.8	44.90	13.75	155.6	142.2	1.366	1.327	0.812
March	171.2	60.71	16.86	199.8	182.7	1.726	1.675	0.798
April	198.1	68.70	19.80	205.1	187.4	1.747	1.696	0.787
May	228.8	75.19	23.63	214.5	196.1	1.801	1.747	0.776
June	235.2	68.03	26.15	209.6	191.5	1.741	1.689	0.768
July	238.3	68.71	27.96	217.2	198.7	1.793	1.739	0.762
August	221.7	64.30	27.76	219.5	201.0	1.814	1.760	0.764
September	183.8	57.20	25.67	205.0	187.3	1.704	1.653	0.768
October	152.8	50.97	22.64	195.1	178.5	1.650	1.602	0.782
November	113.4	37.87	17.99	160.6	147.4	1.390	1.350	0.801
December	99.4	36.31	14.43	149.5	136.6	1.311	1.274	0.811
Year	2067.2	670.54	20.82	2288.0	2091.9	19.421	18.851	0.785

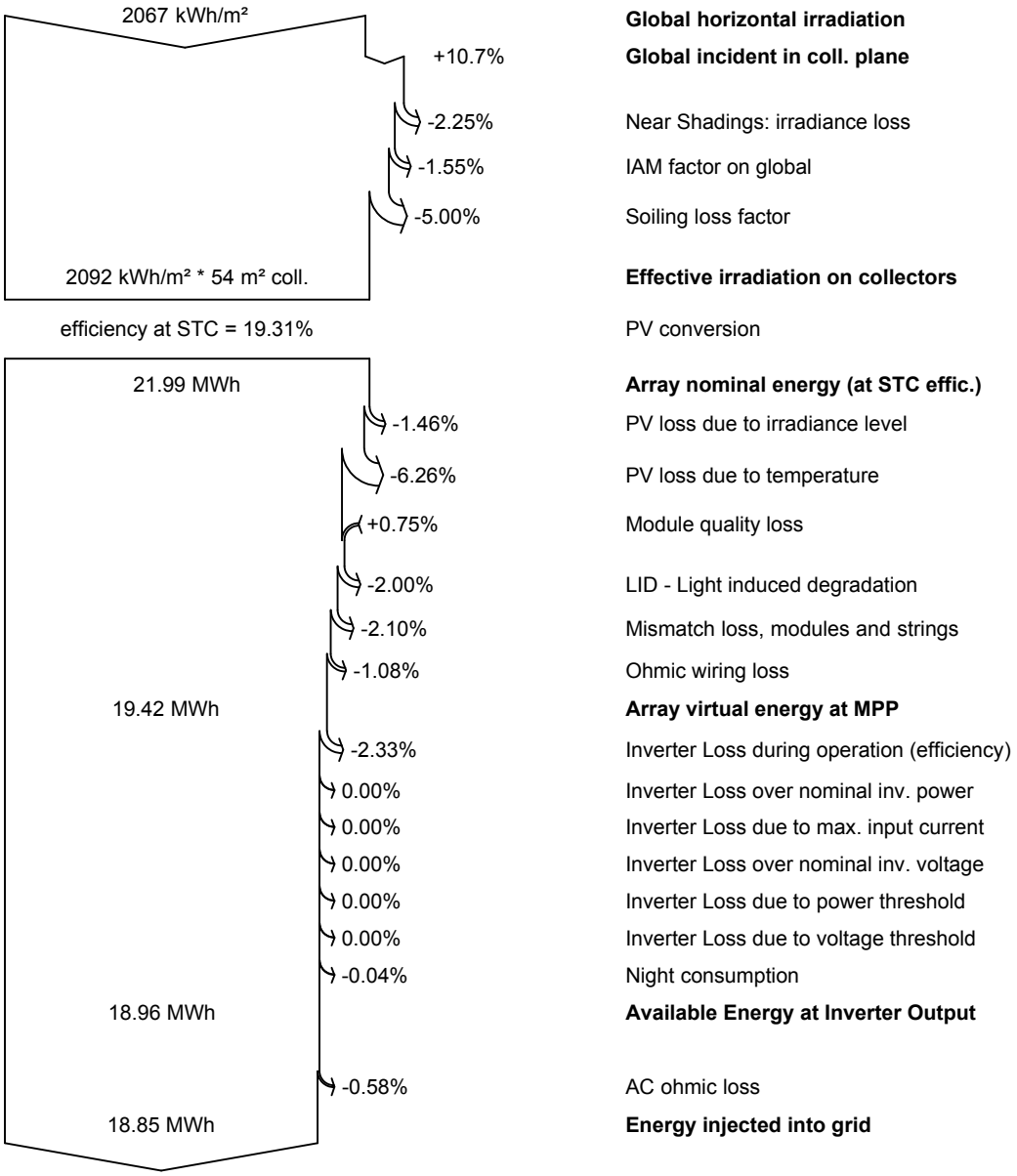
Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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Loss diagram

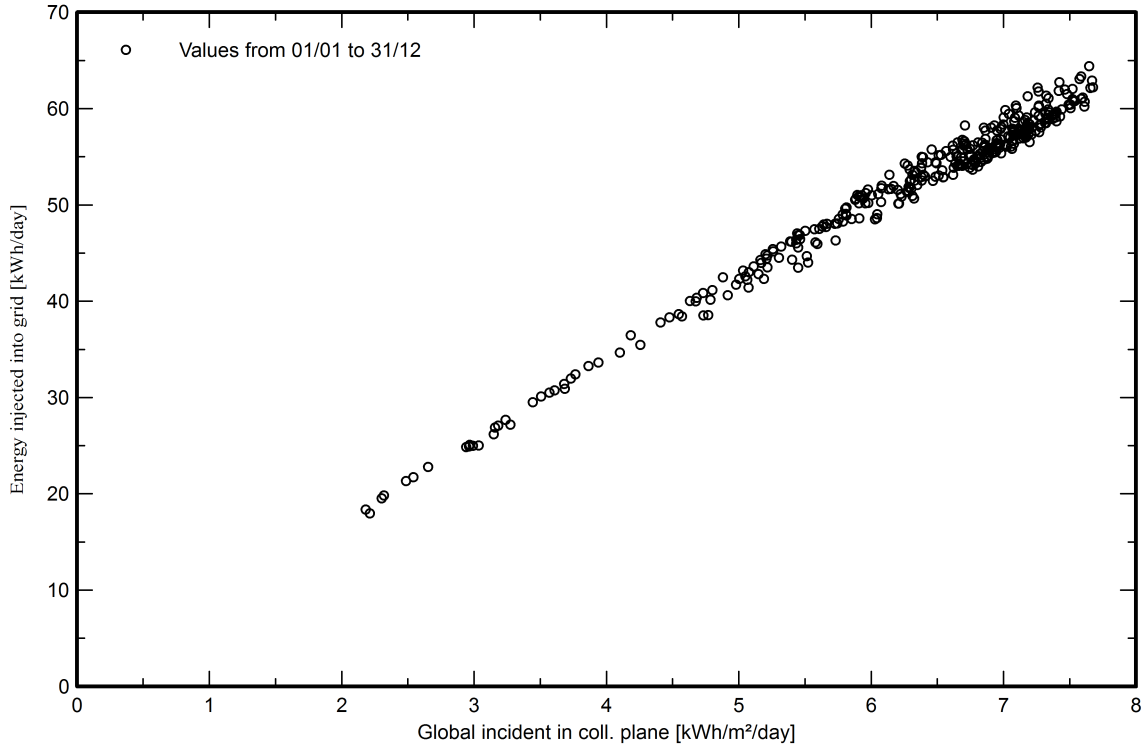


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Special graphs

Daily Input/Output diagram



System Output Power Distribution

